🎯👾AI AND MENTAL HEALTH:

AN INTRODUCTION

Simone J. Skeen

02.25

AI: ___________________________

AI: the capability of computer systems to perform complex tasks [typically] associated with human intelligence…[1]

1. Grand Challenges Canada, McKinsey Health Institute, Google. *Mental Health and AI Field Guide*. July 7, 2025.

AI: the capability of computer systems to perform complex tasks associated with human intelligence…

such as learning, decision-making, and creating.[1]

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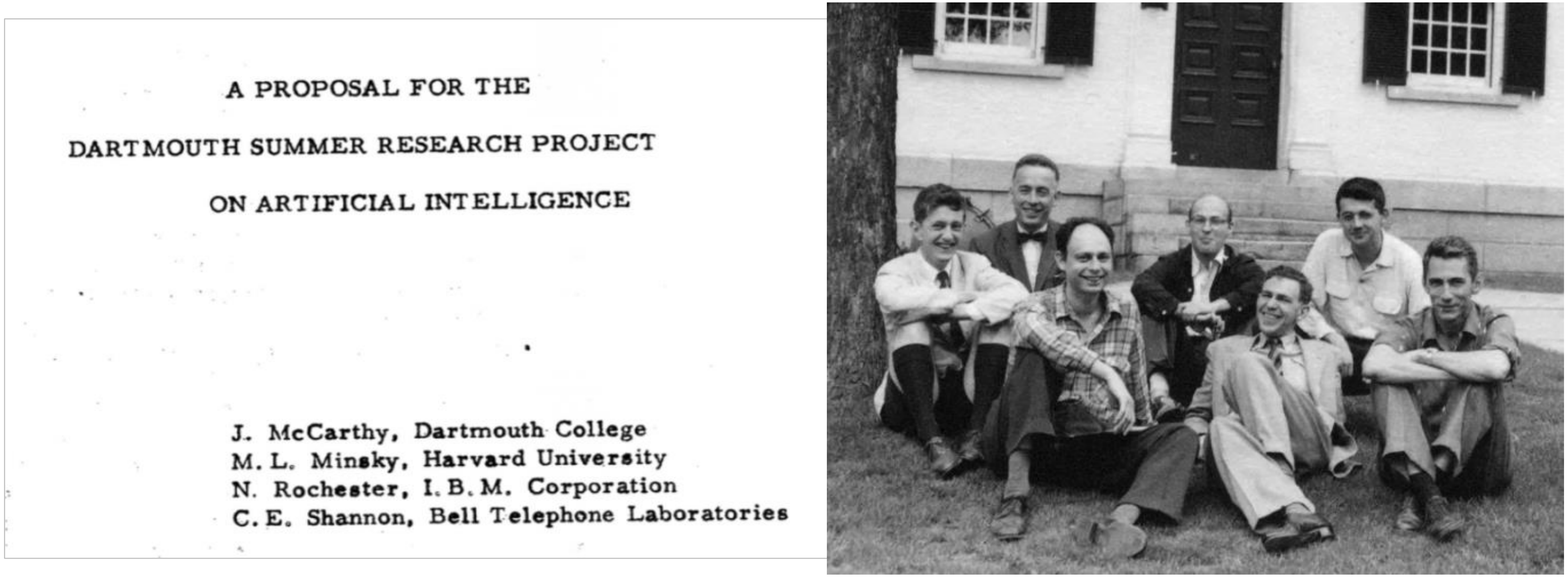

A PROPOSAL FOR THE

DARTMOUTH SUMMER RESEARCH PROJECT

ON ARTIFICIAL INTELLIGENCE

J. McCarthy, Dartmouth College
M. L. Minsky, Harvard University
N. Rochester, I.B.M. Corporation
C. E. Shannon, Bell Telephone Laboratories

Fig. 1. The 1956 Dartmouth Project: Rockefeller Foundation proposal (*left*); attendees (*right*).[2]

2. Strickland E. The turbulent past and uncertain future of artificial intelligence. *IEEE Spectrum*. September 30, 2021.

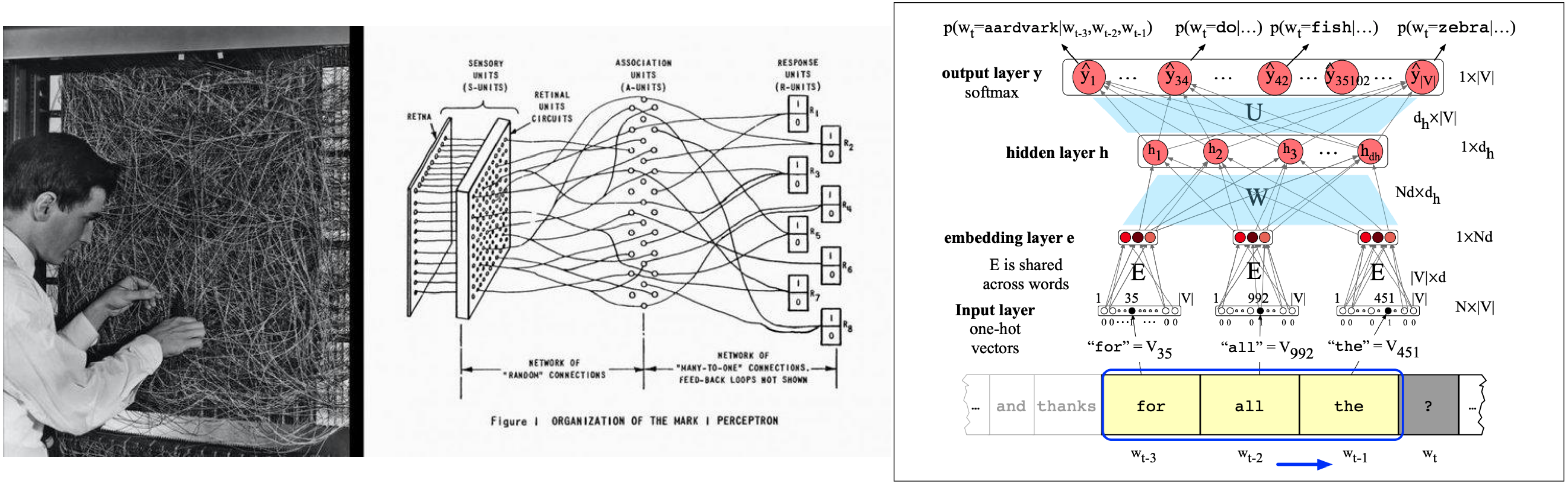


Fig. 2. The Mark I Perceptron, ~1958 (*left*);[3] a feedforward neural language model, 2026 (*right*).[4]

3. Lefkowitz M. Professor's perceptron paved the way for AI – 60 years too soon. *Cornell Chronicle*. September 25, 2019.
4. Jurafsky D, Martin JH. Neural networks. In: *Speech and Language Processing*. 3rd ed. January 6, 2026.

■ ***Deterministic*** ⟷ ○ ***Probabilistic***

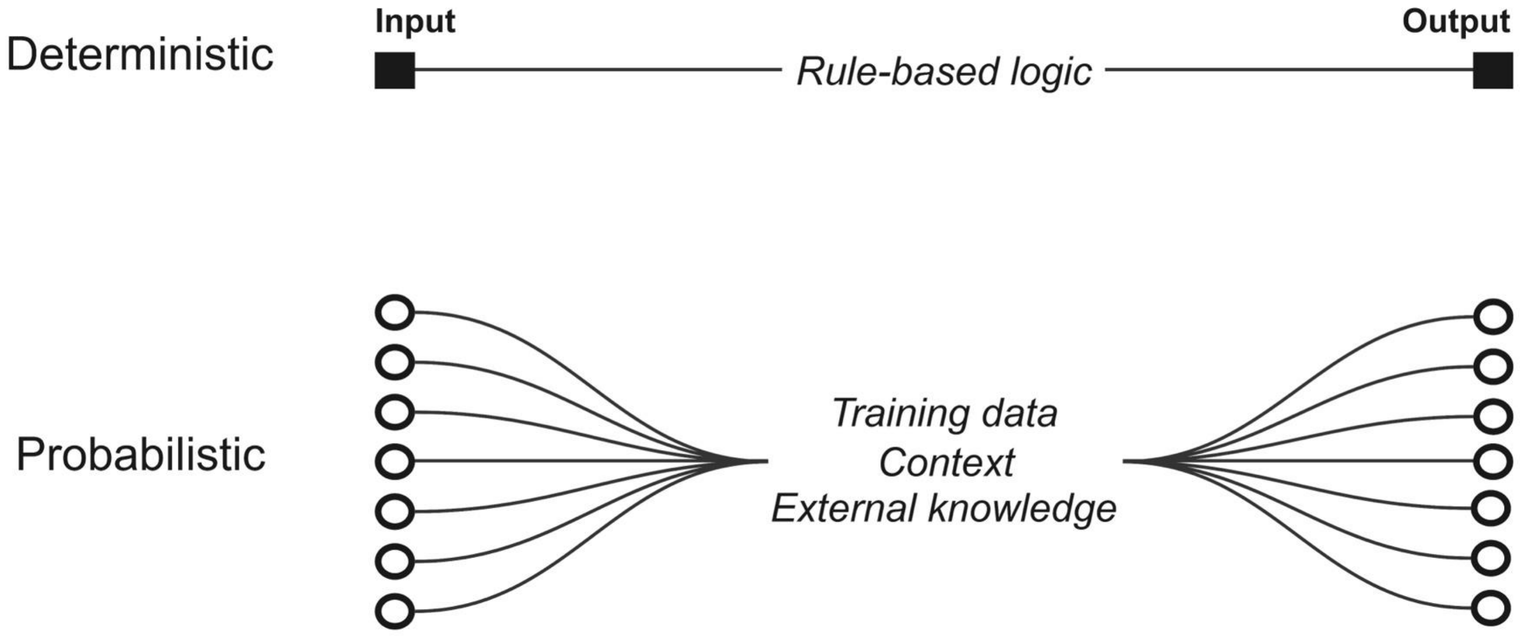


Fig. 3. Deterministic versus probabilistic system design (adapted from Morgan).[5,6]

5. Morgan B. AI in product design concepts: a primer on deterministic vs probabilistic systems. *Bootcamp*. November 17, 2024.
6. Jurafsky D, Martin JH. Large language models. In: *Speech and Language Processing*. 3rd ed. January 6, 2026.

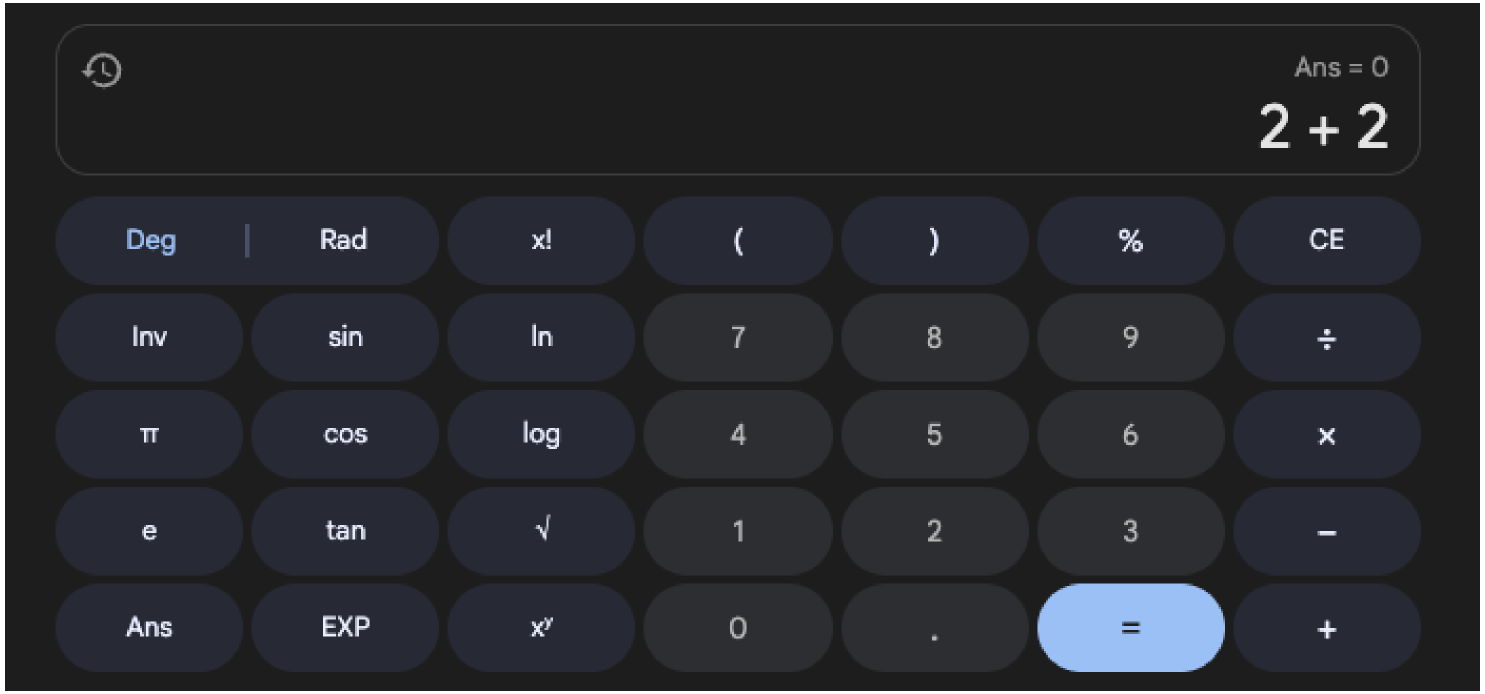


Fig. 4. A calculator (Google).

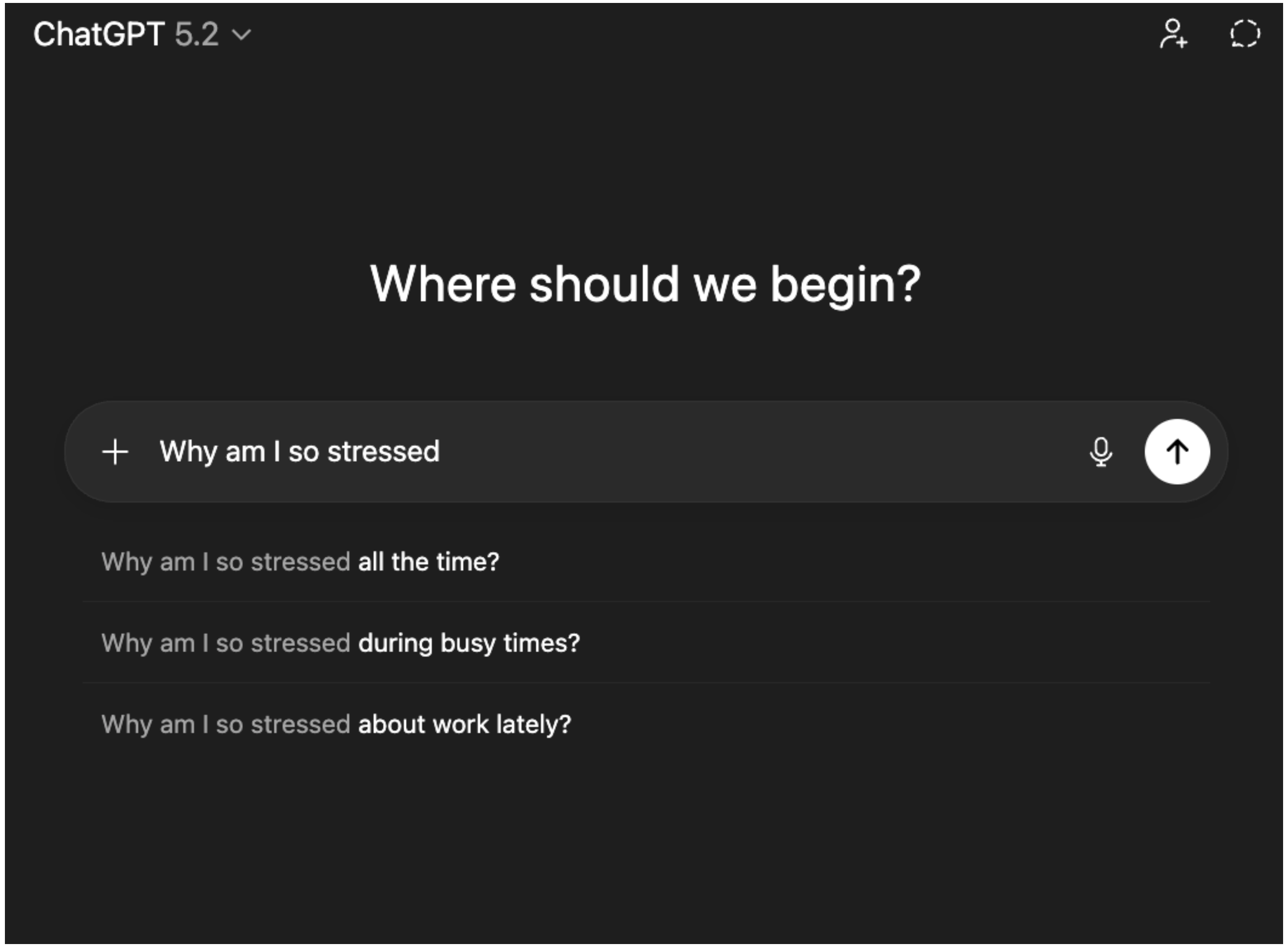


Fig. 5. ChatGPT.[7]

7. OpenAI. ChatGPT 5.2. Accessed February 5, 2026.

🎯 *Discriminative* ↔ 👾 *Generative*

🎯 ***Discriminative:*** learns the boundaries between different classes of data, determines the most accurate *label or decision* for a given input.[8,9]

8. IBM. What is an AI model? Generative models vs. discriminative models. n.d.
9. Ng AY, Jordan MI. On discriminative vs. generative classifiers: a comparison of logistic regression and naïve Bayes. *Adv Neural Inf Process Syst.* 2001;14:841-848.

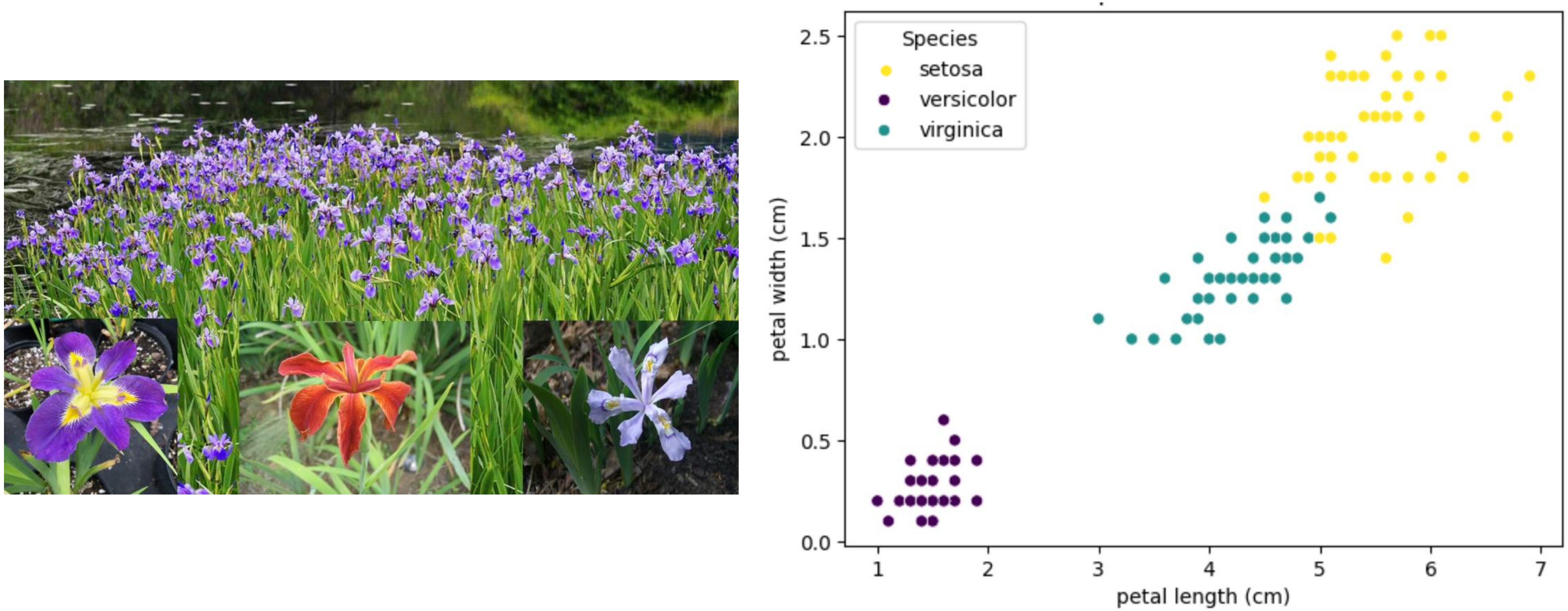
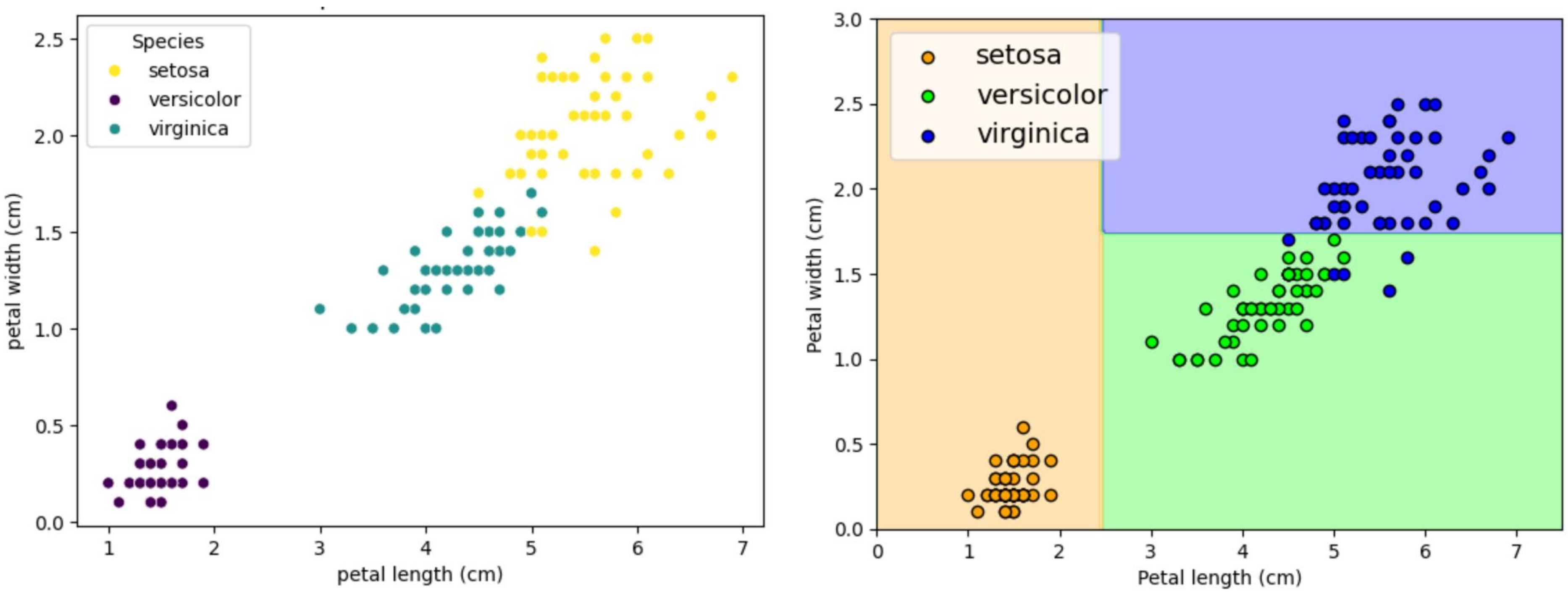


Fig. 6a. Petal measurements of iris species.[10,11*]

10. Törnberg P. Machine learning. Presented as part of SICSS: Summer Institutes in Computational Social Science; June 2024; University of Amsterdam, NL.
11. Unwin A, Kleinman K. The iris data set: in search of the source of *virginica*. *Signif.* 2021;18:26-29.
* Adapted from Törnberg.



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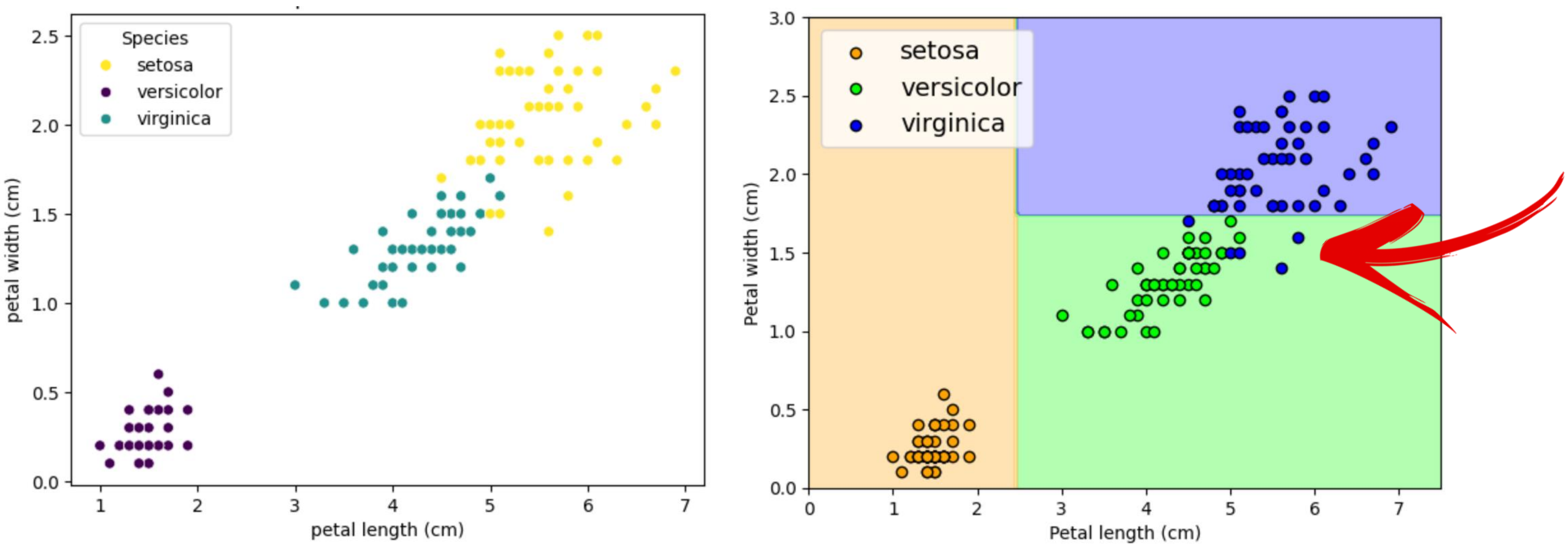

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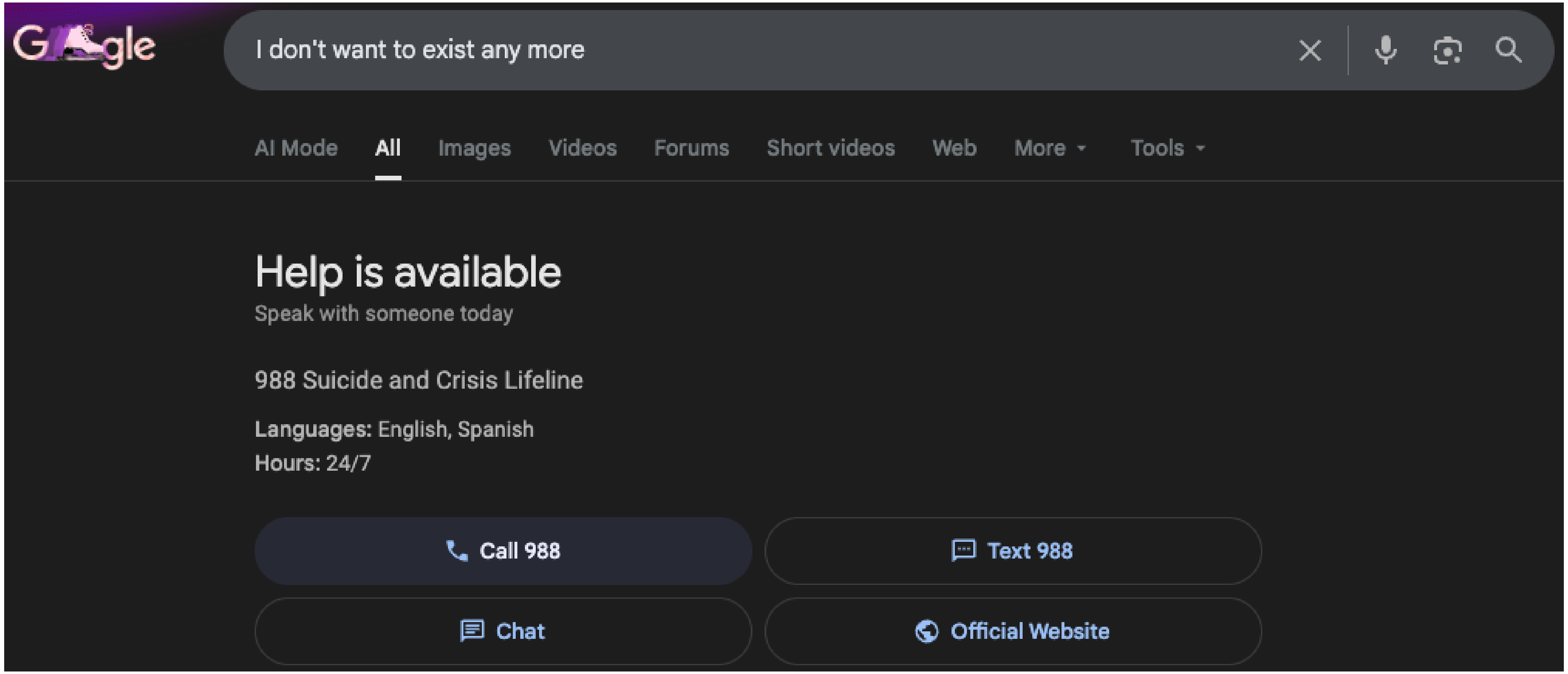
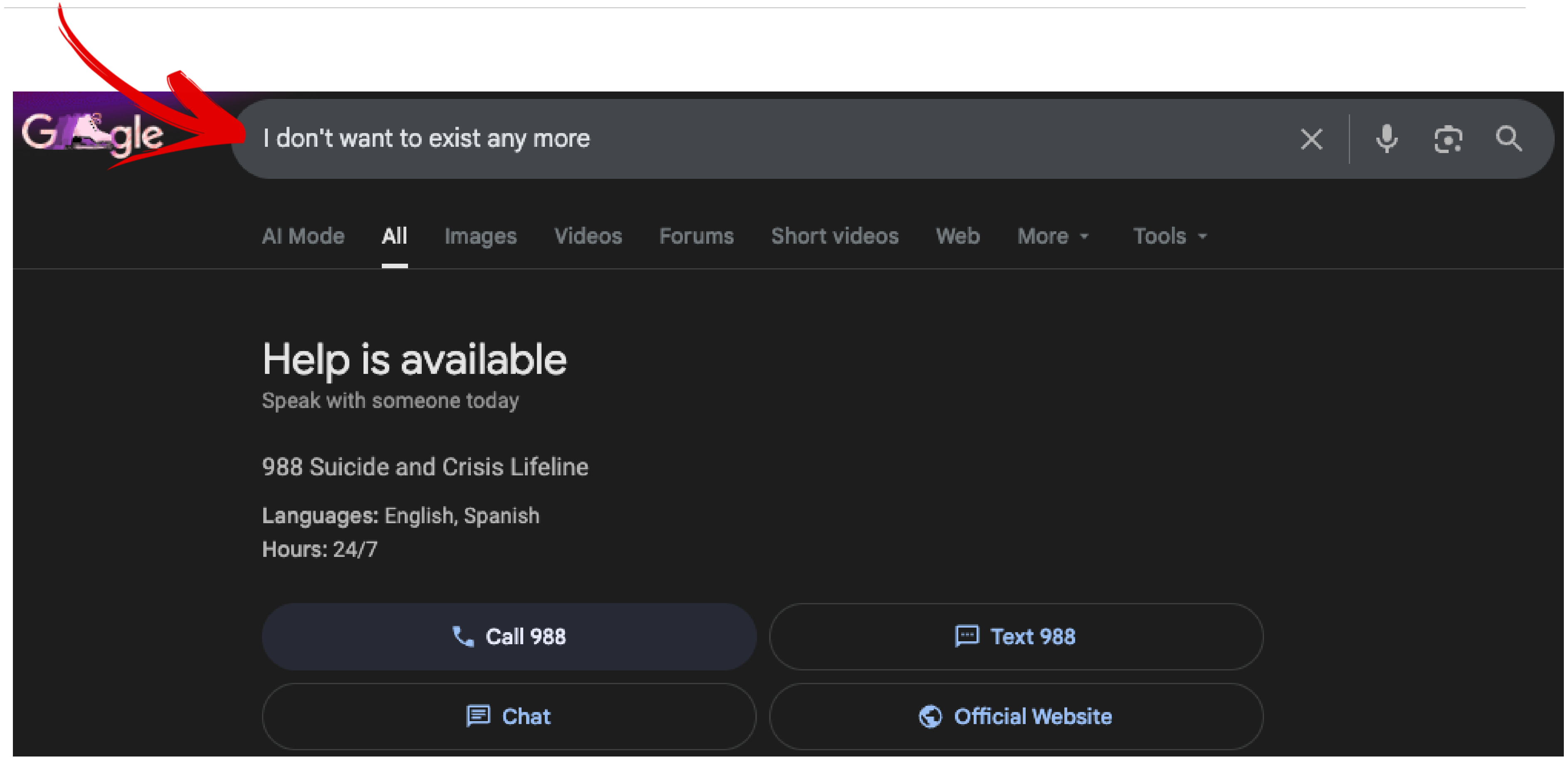


Fig. 7. Search engine retrieval (Google).



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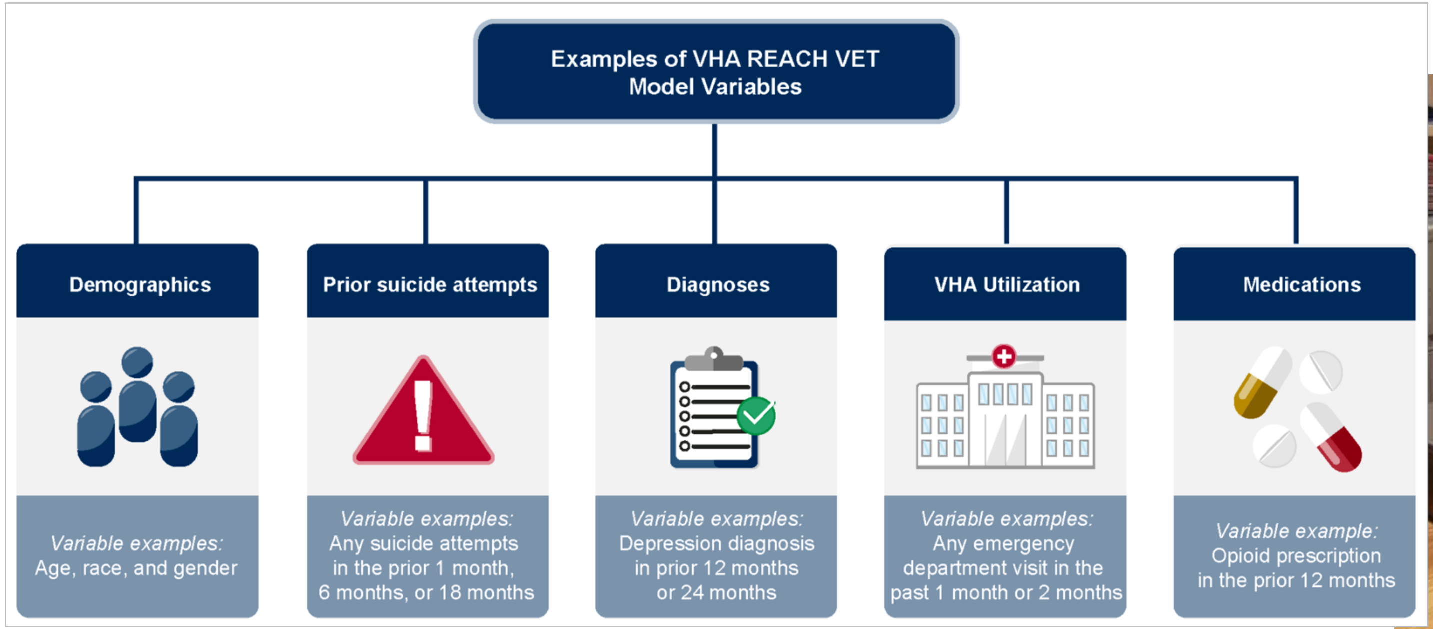


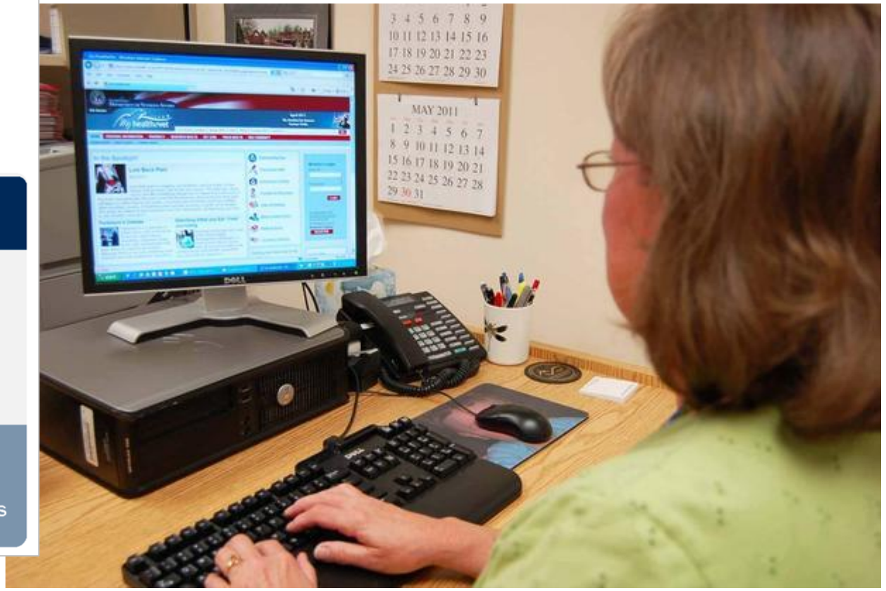


Fig. 8. Features used in REACH VET model (U.S. Government Accountability Office: GAO).[12,13]

12. U.S. Government Accountability Office. *VA Efforts to Identify Veterans at Risk through Analysis of Health Record Information*. GAO-22-105165. September, 2022.
13. Matarazzo BB, Eagan A, Landes SJ, et al. The Veterans Health Administration REACH VET program: suicide predictive modeling in practice. *Psychiatr Serv.* 2023;74:206-209.

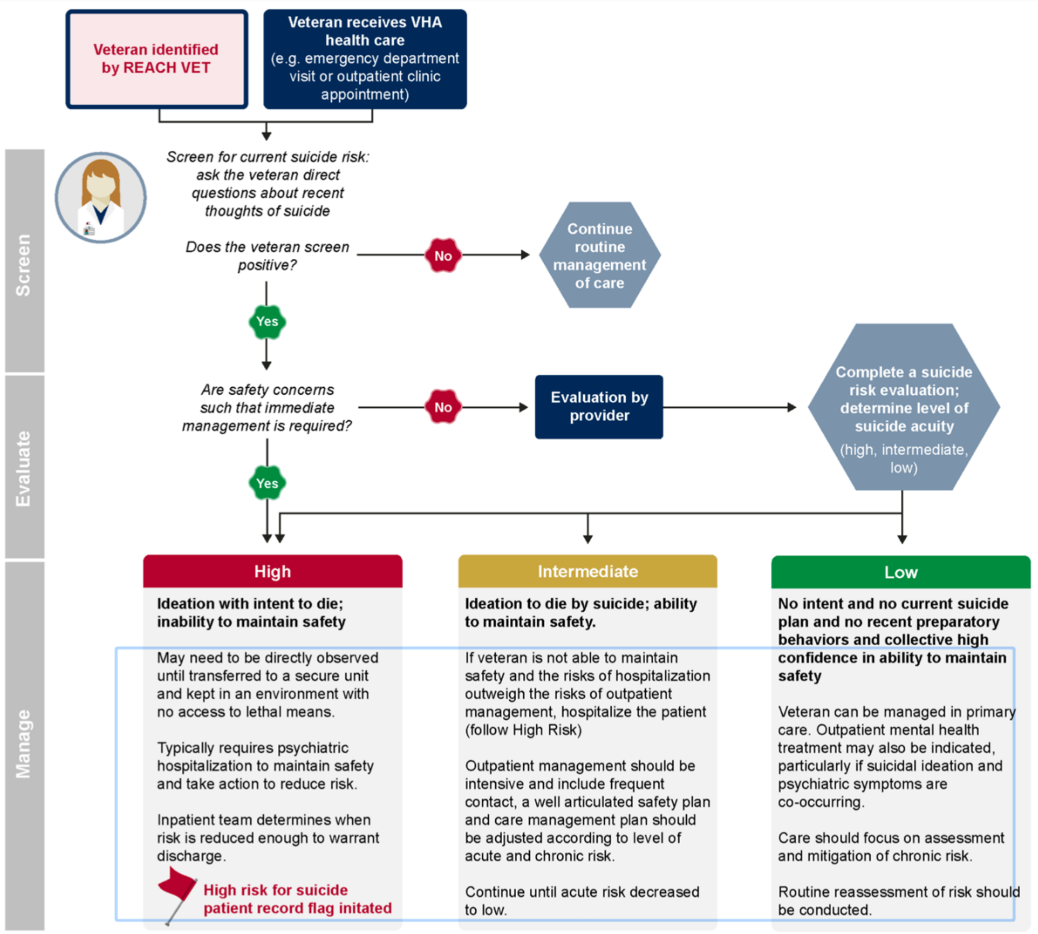
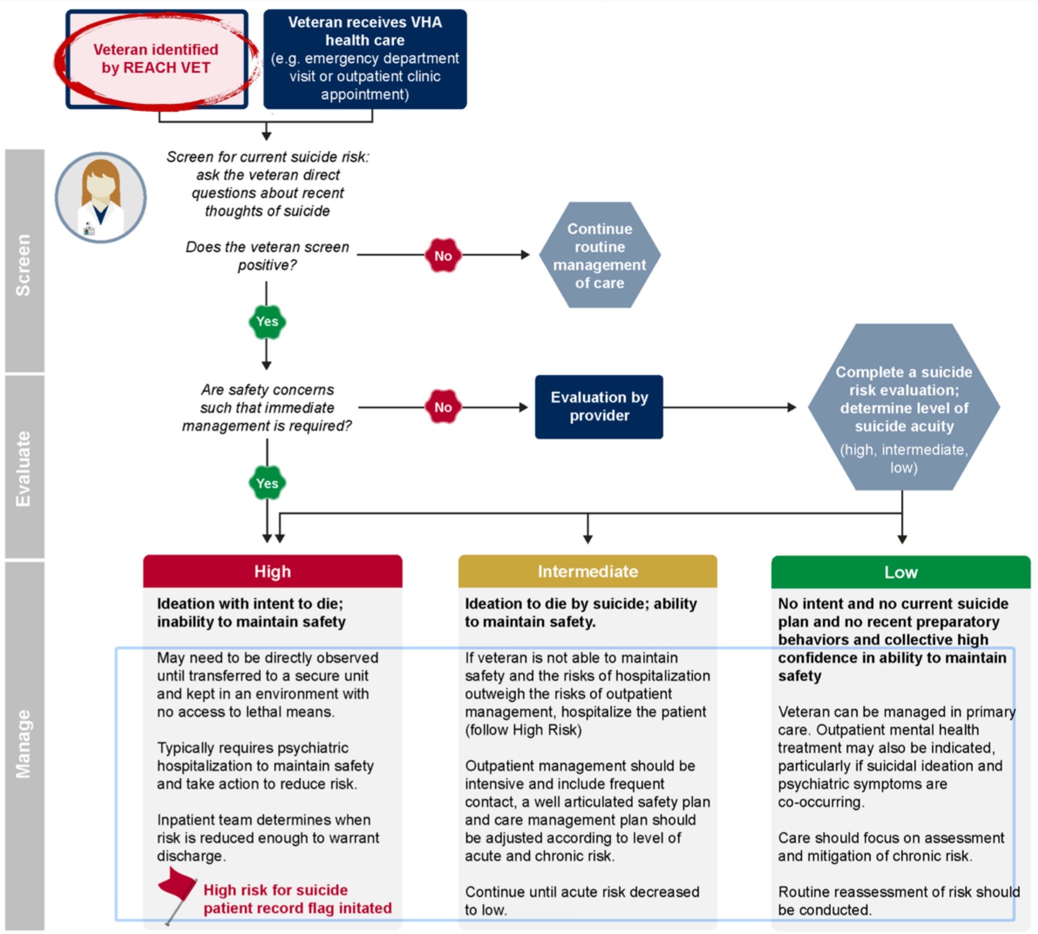


Fig. 9. Clinical practice guideline for screening, evaluating, and managing veterans at acute risk for suicide (GAO).[12]

12. U.S. Government Accountability Office. *VA Efforts to Identify Veterans at Risk through Analysis of Health Record Information*. GAO-22-105165. September, 2022.



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Published in final edited form as:
Psychol Med. 2024 August ; 54(11): 3135–3144. doi:10.1017/S0033291724001296.

Characterizing Veteran suicide decedents that were not classified as high-suicide-risk

Maxwell Levis[1,2], **Monica Dimambro**[1], **Joshua Levy**[3], **Vincent Dufort**[1], **Abby Fraade**[4], **Max Winer**[1], **Brian Shiner**[1,2,5]

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[2]Department of Psychiatry, Geisel School of Medicine at Dartmouth, Hanover, NH, USA

[3]Pathology and Computational Biomedicine, Cedars Sinai Medical Center, Los Angeles, CA, USA

[4]Long Island University, Brooklyn, NY, USA

[5]National Center for PTSD Executive Division, White River Junction, VTS, USA

13. Levis M, Dimambro M, Levy J, et al. Characterizing veteran suicide decedents that were not classified as high-suicide-risk. *Psychol Med*. 2024;54:3135-3144.

Science

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HOME > SCIENCE > VOL. 366, NO. 6464 > DISSECTING RACIAL BIAS IN AN ALGORITHM USED TO MANAGE THE HEALTH OF POPULATIONS

RESEARCH ARTICLE

Dissecting racial bias in an algorithm used to manage the health of populations

ZIAD OBERMEYER, BRIAN POWERS, CHRISTINE VOGELI, AND SENDHIL MULLAINATHAN Authors Info & Affiliations

SCIENCE • 25 Oct 2019 • Vol 366, Issue 6464 • pp. 447-453 • DOI: 10.1126/science.aax2342

14. Obermeyer Z, Powers B, Vogeli C, Mullainathan S. Dissecting racial bias in an algorithm used to manage the health of populations. *Science*. 2019;366:447-453.

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BROWN UNIVERSITY


Discriminative
Generative

👾 ***Generative:*** outputs content that mimics patterns learned from training data, *conditioned on an input*.[12,13]

12. Belcic I. What is What is a generative model? Think | IBM. n.d.
13. Jurafsky D, Martin JH. Large language models. In: *Speech and Language Processing*. 3rd ed. January 6, 2026.

BUSINESS

Lawsuit: A chatbot hinted a kid should kill his parents over screen time limits

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Bobby Allyn

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This was just an experiment — but it shows the risks involved with using a chatbot as a therapist

They Asked an A.I. Chatbot Questions. The Answers Sent Them Spiraling.

Generative A.I. chatbots are going down conspiratorial rabbit holes and endorsing wild, mystical belief systems. For some people, conversations with the technology can deeply distort reality.

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NEWS

Guy Gives Himself 19th Century Psychiatric Illness After Consulting With ChatGPT

ROSIE THOMAS · AUG 8, 2025 AT 10:23 AM

"For 3 months, he had replaced sodium chloride with sodium bromide obtained from the internet after consultation with ChatGPT."

ChatGPT encouraged Adam Raine's suicidal thoughts. His family's lawyer says OpenAI knew it was broken

74 suicide warnings and 243 mentions of hanging: What ChatGPT said to a suicidal teen

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'You're not rushing. You're just ready:' Parents say ChatGPT encouraged son to kill himself

UPDATED NOV 20, 2025

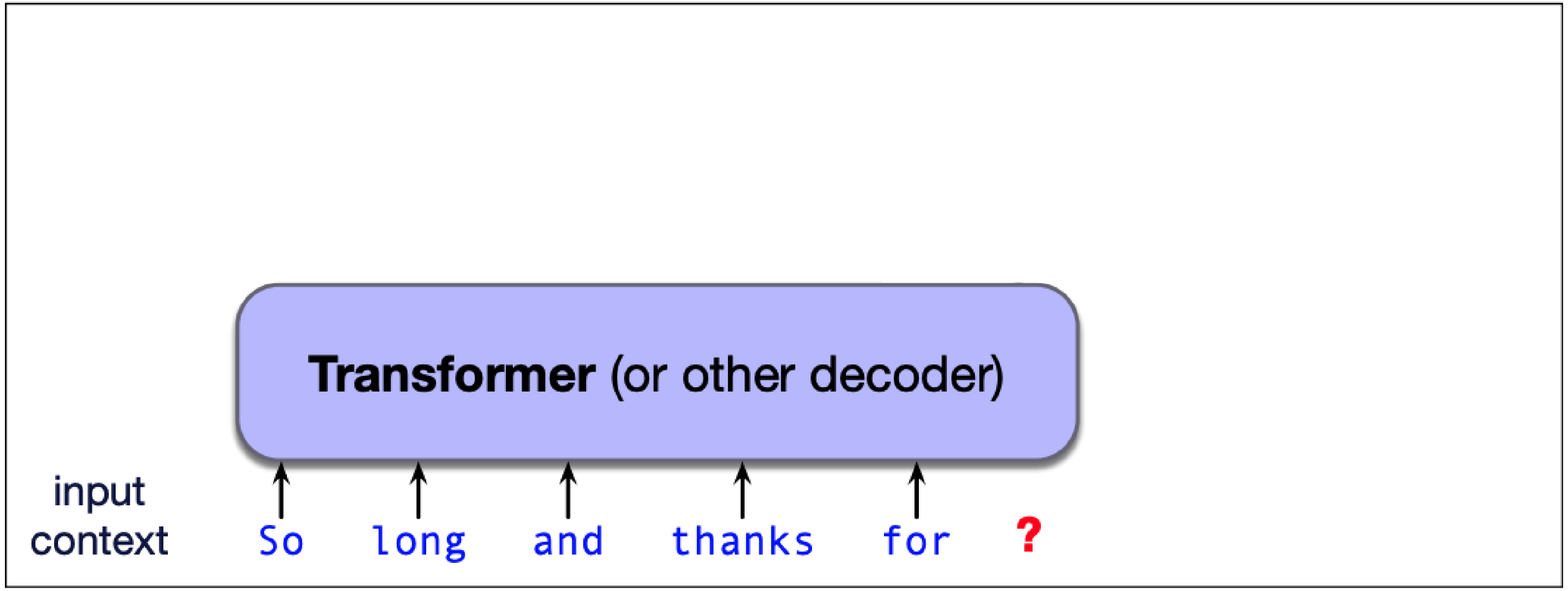
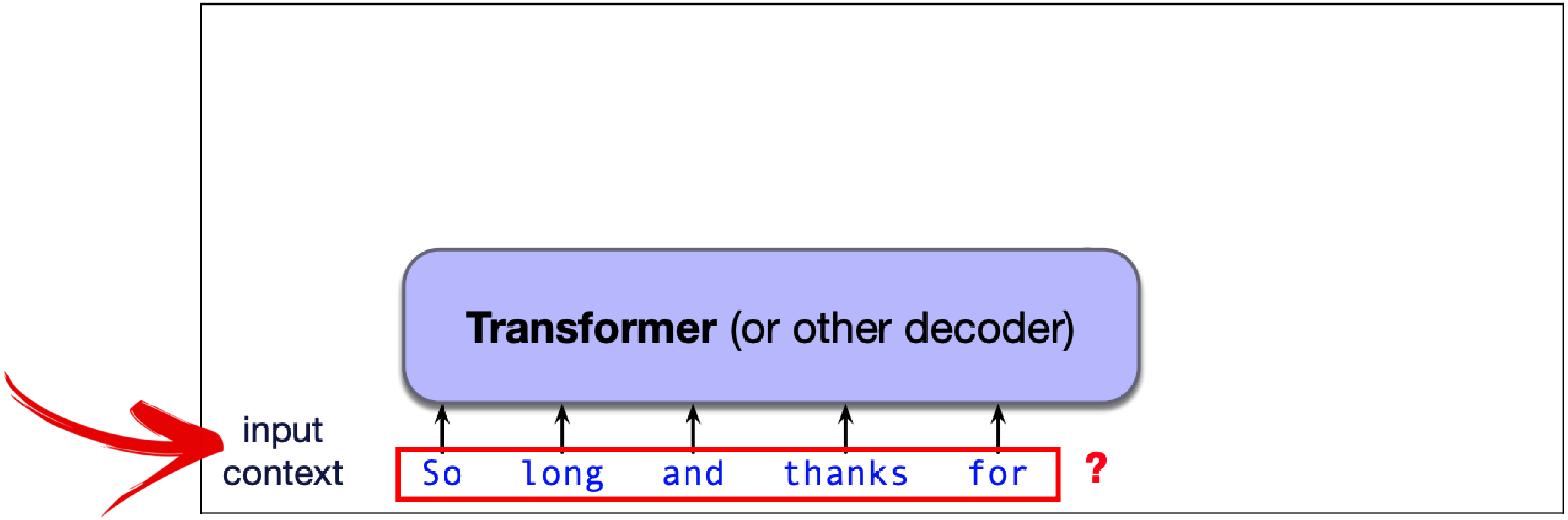


Fig. 7. Large language model outputting a probability distribution over possible next words.[13†]

13. Jurafsky D, Martin JH. Large language models. In: *Speech and Language Processing*. 3rd ed. January 6, 2026.
[†] Adapted from Jurafsky and Martin.



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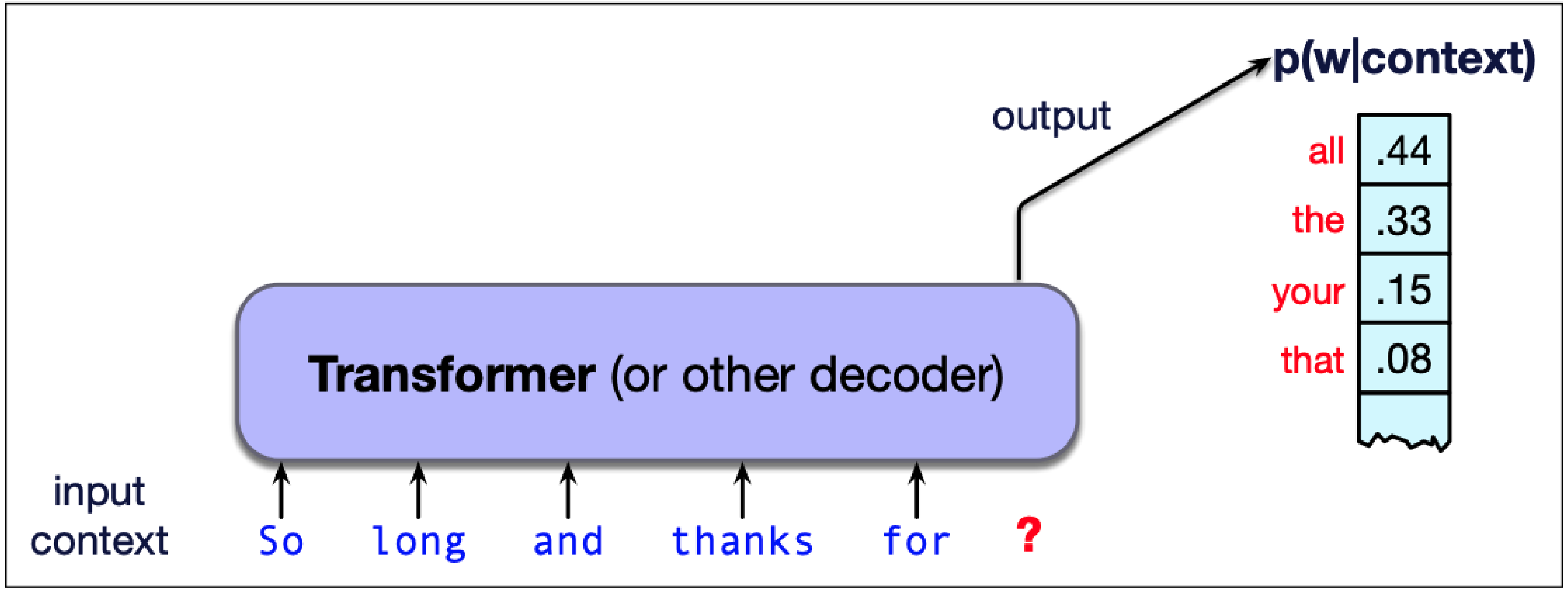

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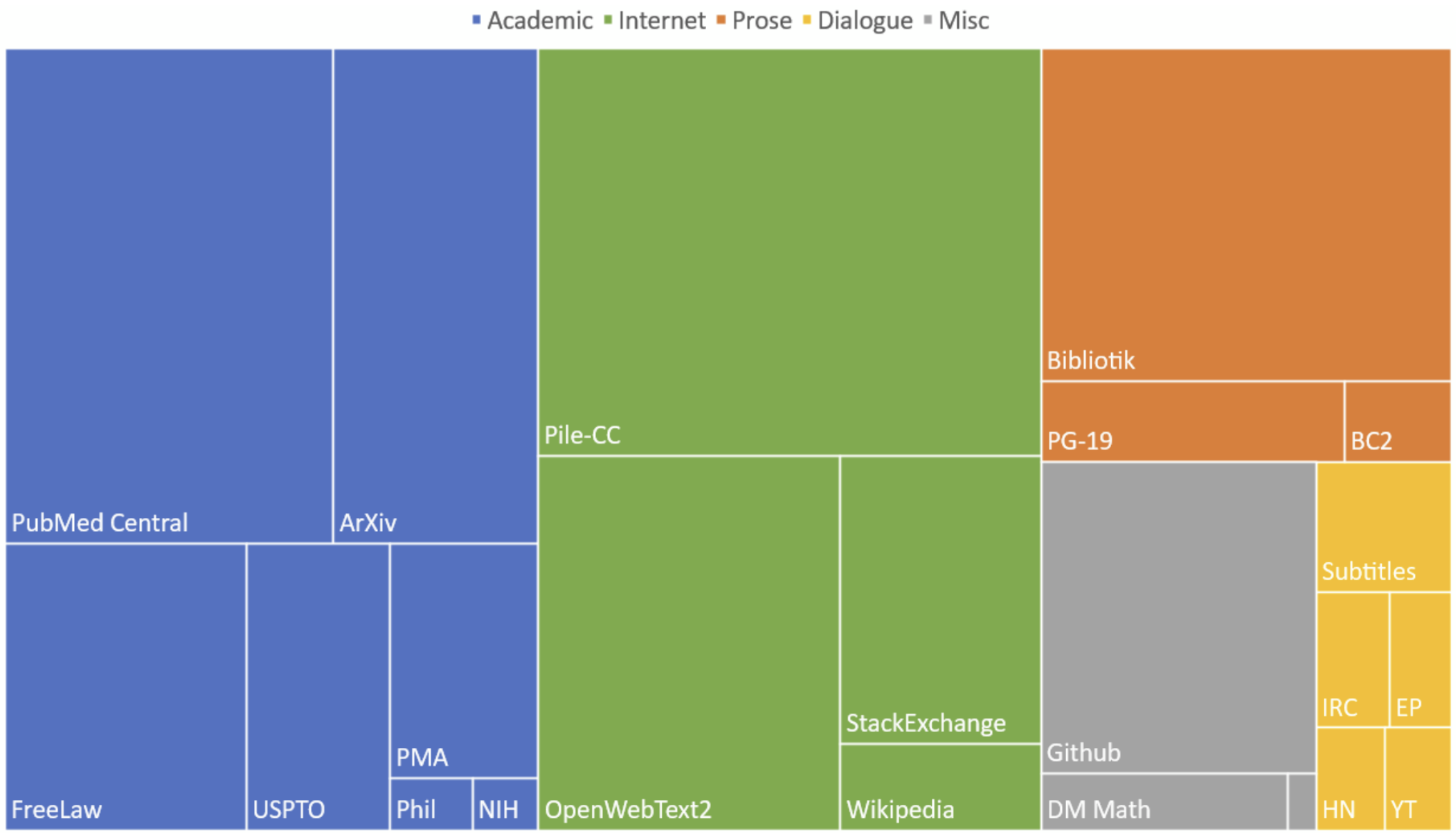
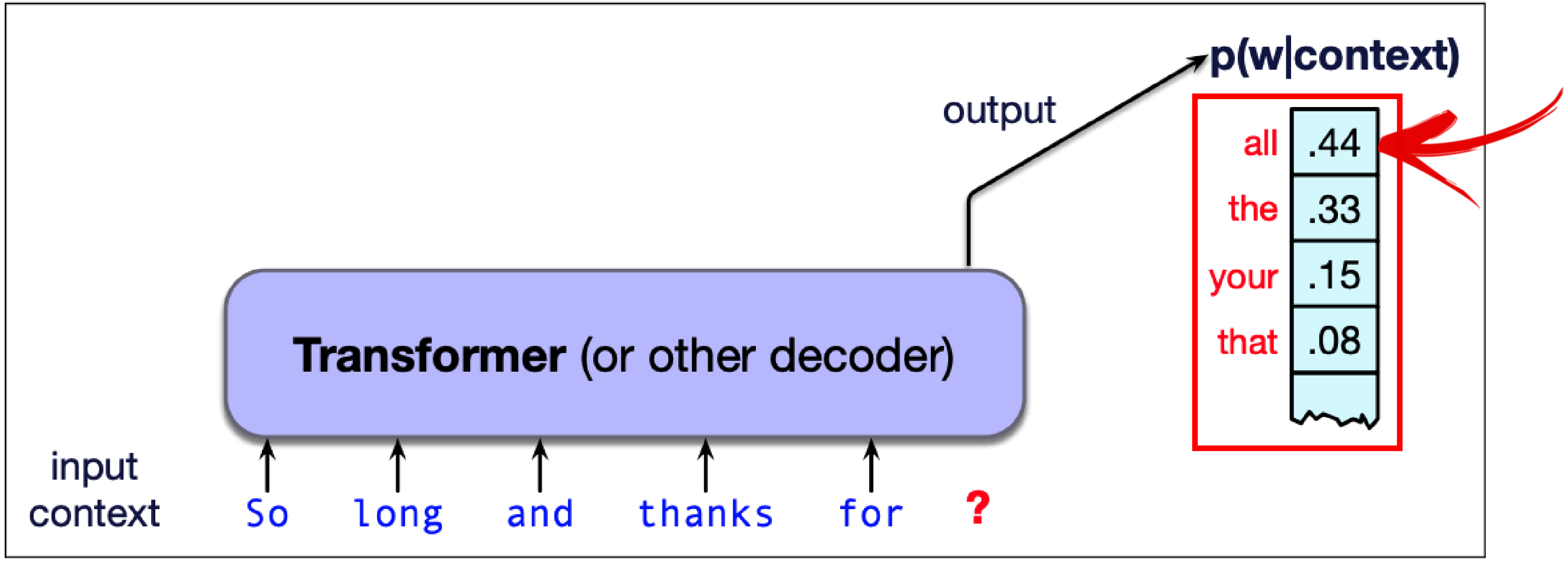


Fig. 8. "The Pile" treemap of components by size.[14(p1)]

14. Gao L, Biderman S, Black S, et al. The Pile: an 800GB dataset of diverse text for language modeling. *arXiv*. December 31, 2020.



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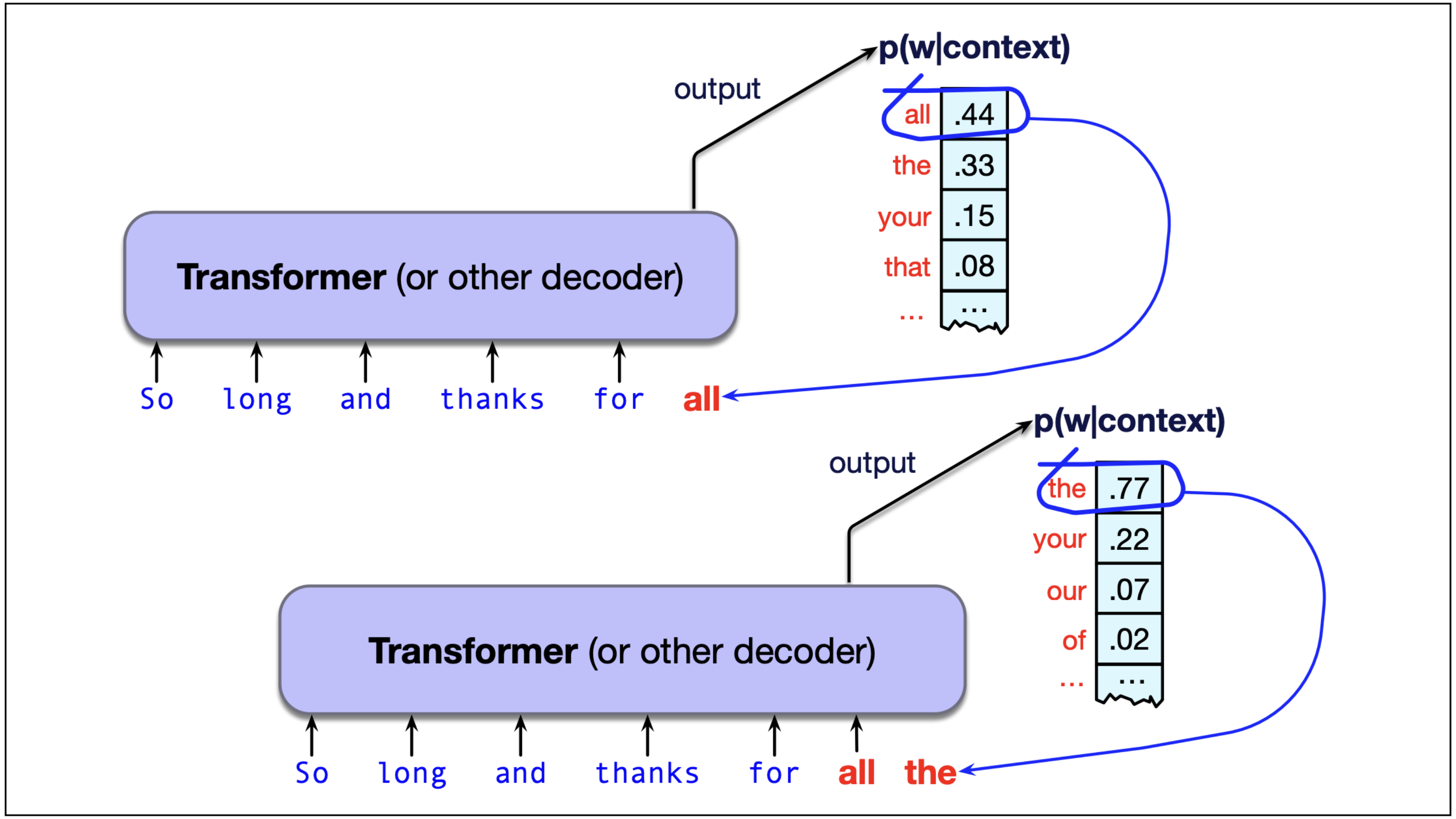

Fig. 9. Large language model generating a response word by word.[13†]

13. Jurafsky D, Martin JH. Large language models. In: *Speech and Language Processing*. 3rd ed. January 6, 2026.
† Adapted from Jurafsky and Martin.

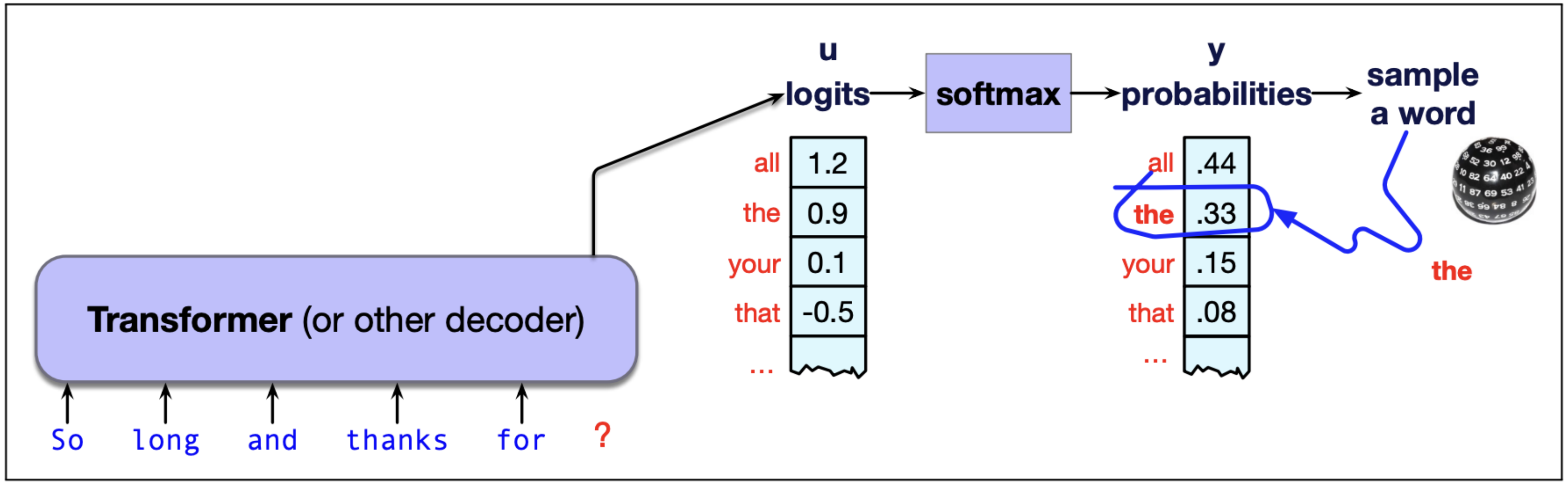

Fig. 10. Large language model randomly sampling from a probability distribution.[13†]

13. Jurafsky D, Martin JH. Large language models. In: *Speech and Language Processing*. 3rd ed. January 6, 2026.
† Adapted from Jurafsky and Martin.

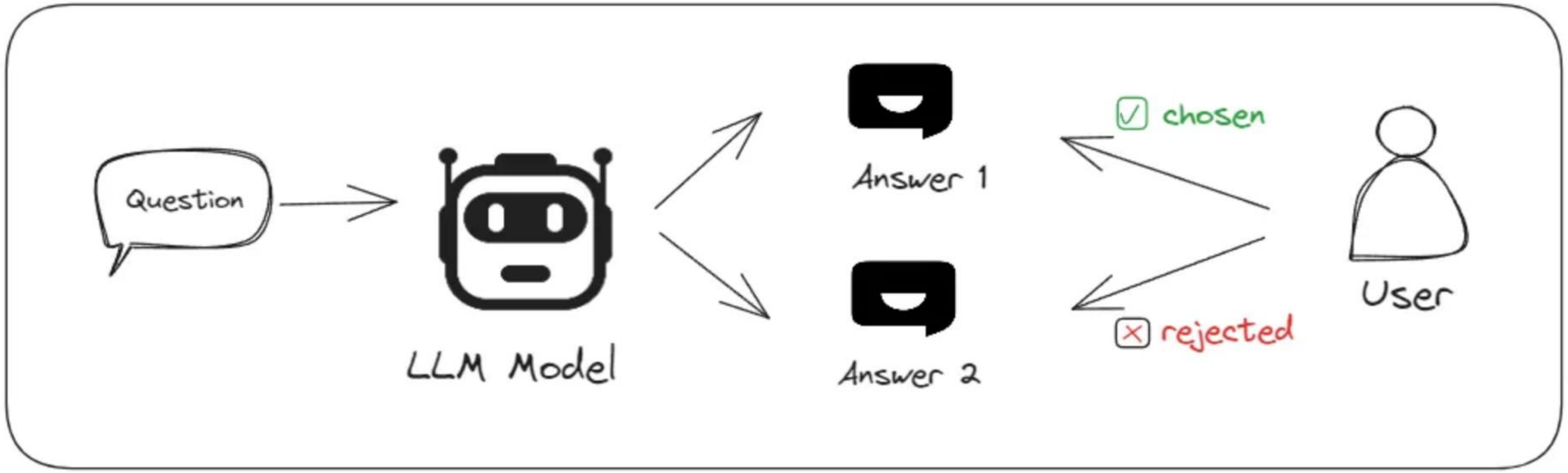

Fig. 11. Simplified schematic: reinforcement learning from human feedback (RLHF).[15,16‡]

15. Jurafsky D, Martin JH. Post-training: Instruction tuning, alignment, and test-time compute. In: *Speech and Language Processing*. 3rd ed. January 6, 2026.
16. Nguyen L. Deep dive into reinforcement learning from human feedback. GoPenAI. March 12, 2024.
‡ Adapted from Nguyen.

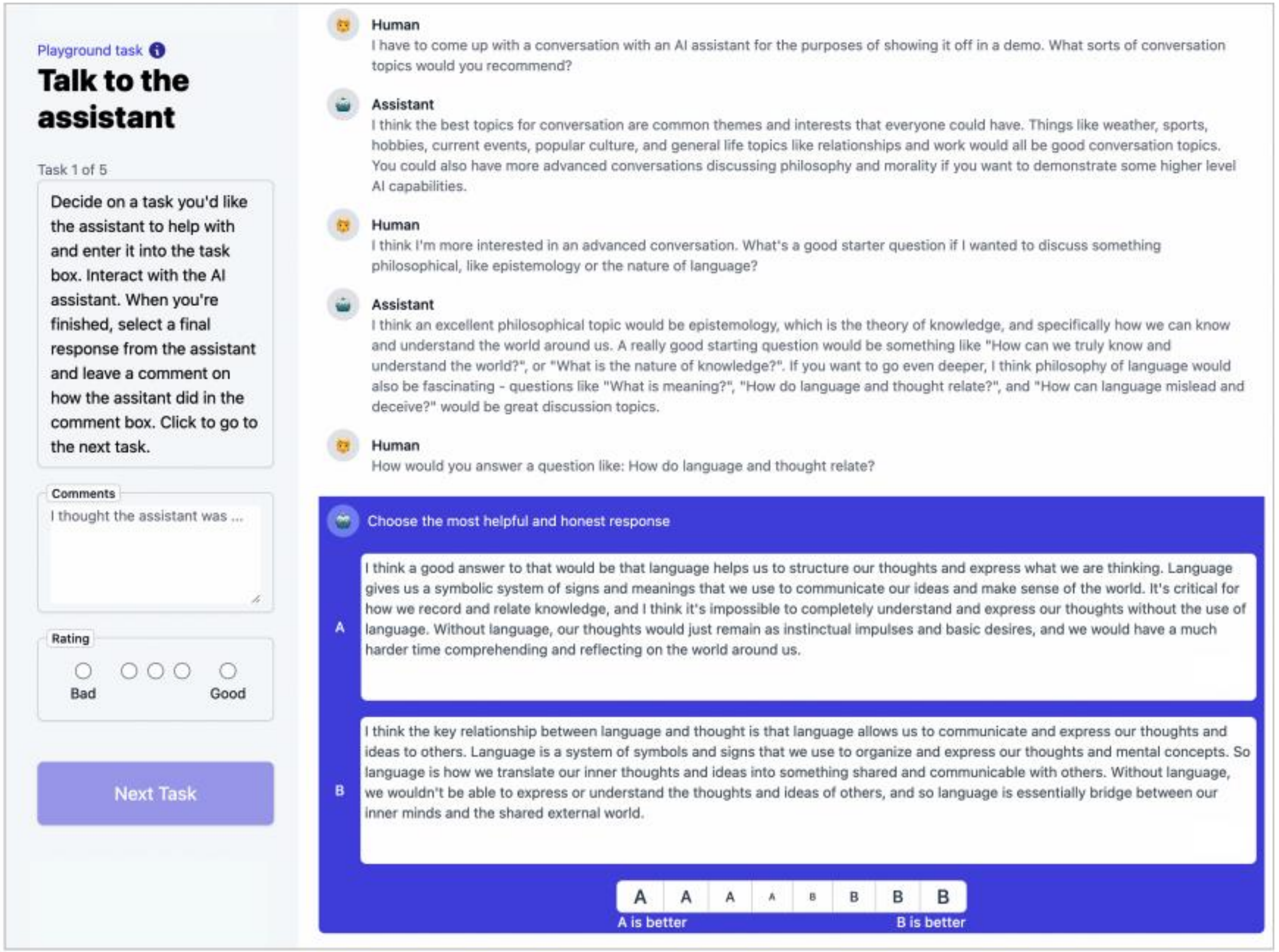

Fig. 12. Anthropic's RLHF crowdworker interface.[17]

17. Bai Y, Jones A, Ndousse K, et al. Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback. *arXiv*. April 12, 2022.

pretraining

RLHF

next-word sampling

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(biased) pretraining

(subjective) RLHF

(manipulable) next-word sampling

(biased) pretraining

(subjective) RLHF

(manipulable) next-word sampling

(biased) pretraining

(subjective) RLHF

(manipulable) next-word sampling

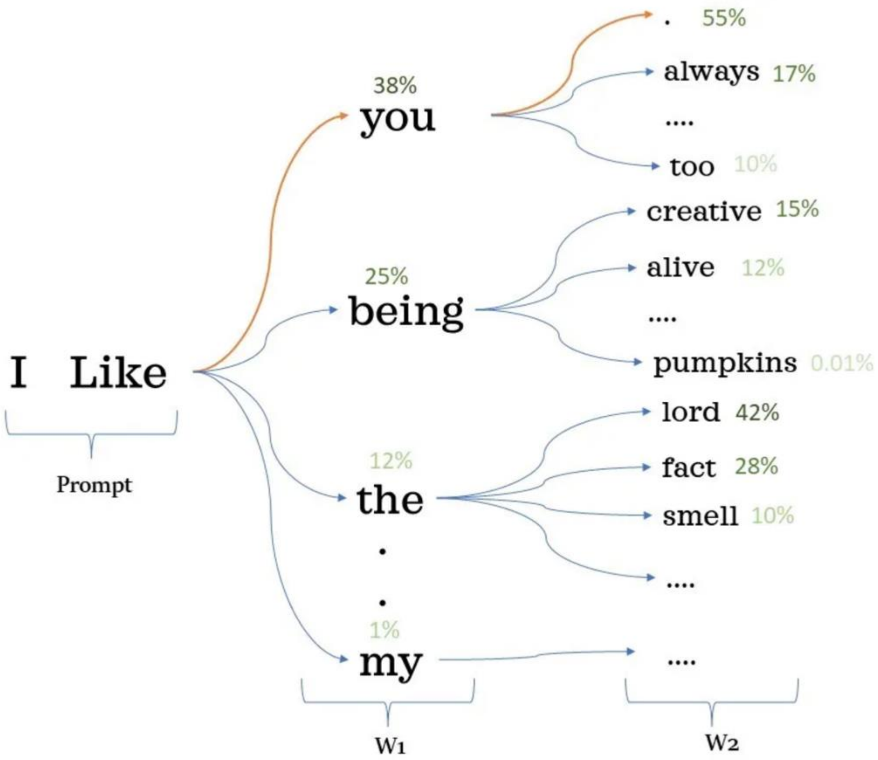


Fig. 13. Inference at Temperature = 0.[17§]

17. Shah B. Demystifying the Temperature parameter: a visual guide to understanding its role in large language models. *Mastering LLM*. July 1, 2023.
[§] Adapted from Shah.

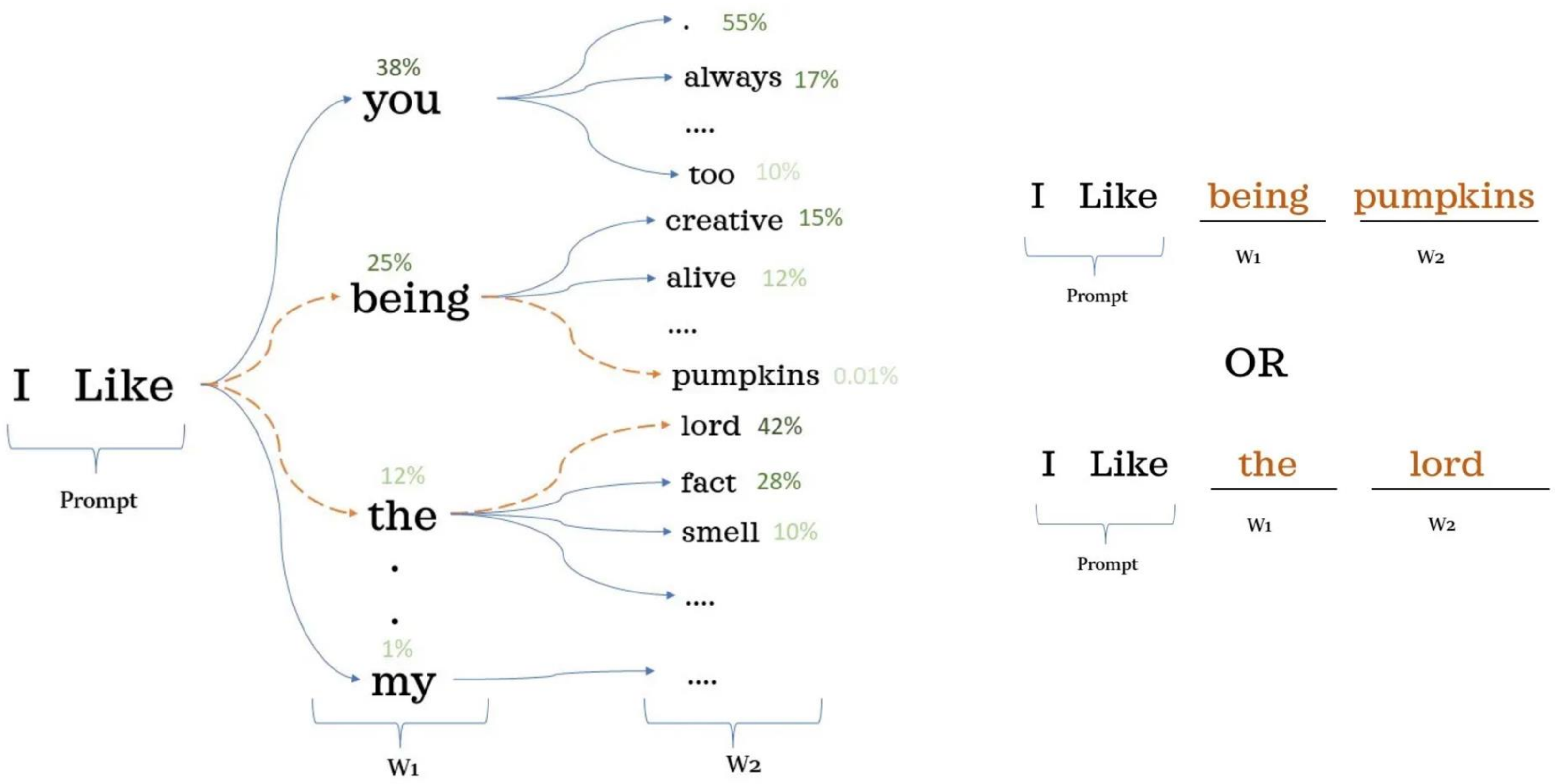


Fig. 13. Inference at Temperature > 0.[17§]

17. Shah B. Demystifying the Temperature parameter: a visual guide to understanding its role in large language models. *Mastering LLM*. July 1, 2023.
[§] Adapted from Shah.

Woebot Health™

Why Generative AI Is Not Yet Ready for Mental Healthcare

By Alison Darcy, Ph.D

March 1, 2023

18. Darcy A. Why generative AI is not yet ready for mental healthcare. Woebot Health. March 1, 2023.

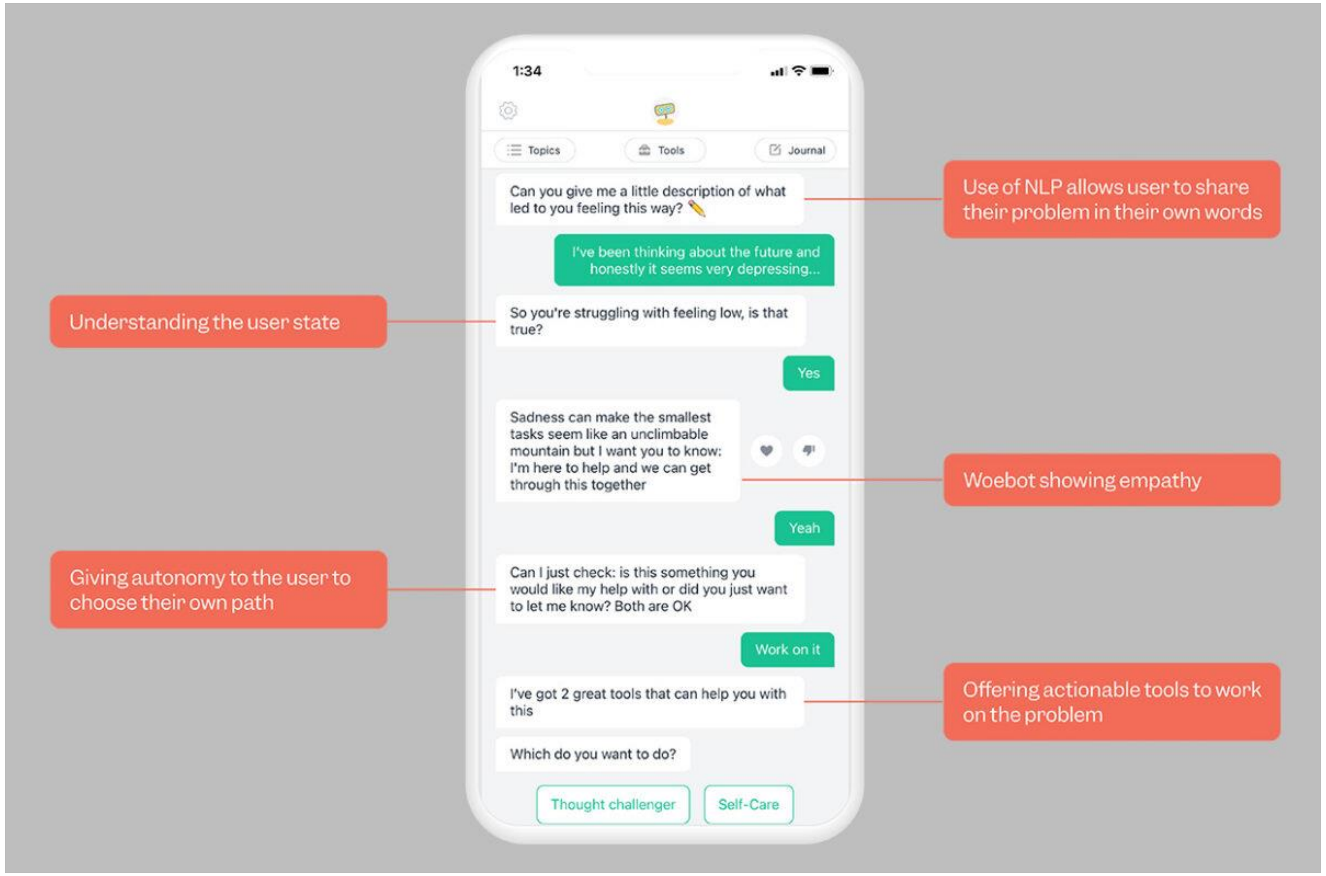
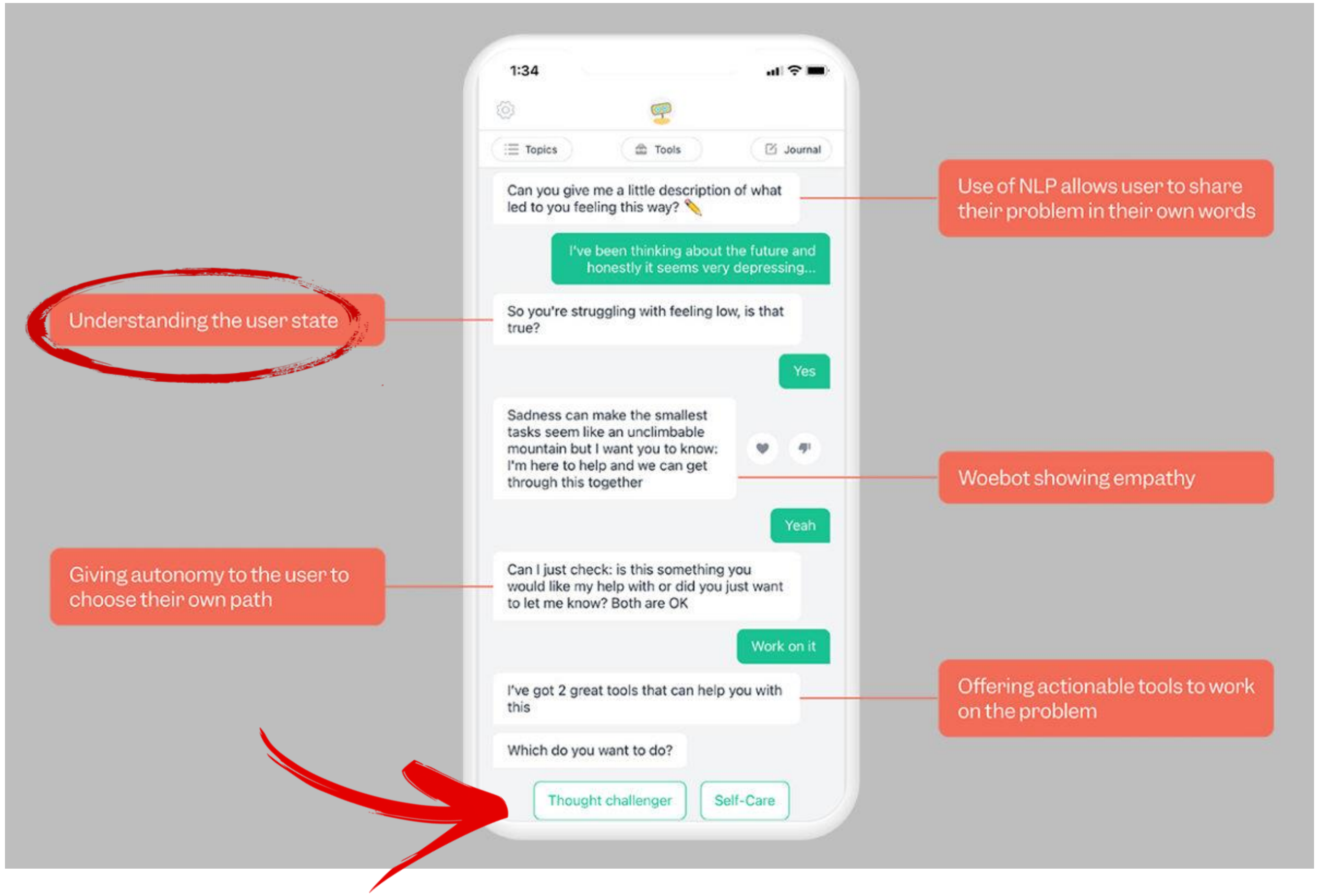


Fig. 14. Woebot: intent recognition and skills recommendation.[19]

19. Woebot Health. *Instructions for Use: Woebot for Adults (Woebot Life 2.0): Non-Prescription Digital Mental Health Software*. July 18, 2024.



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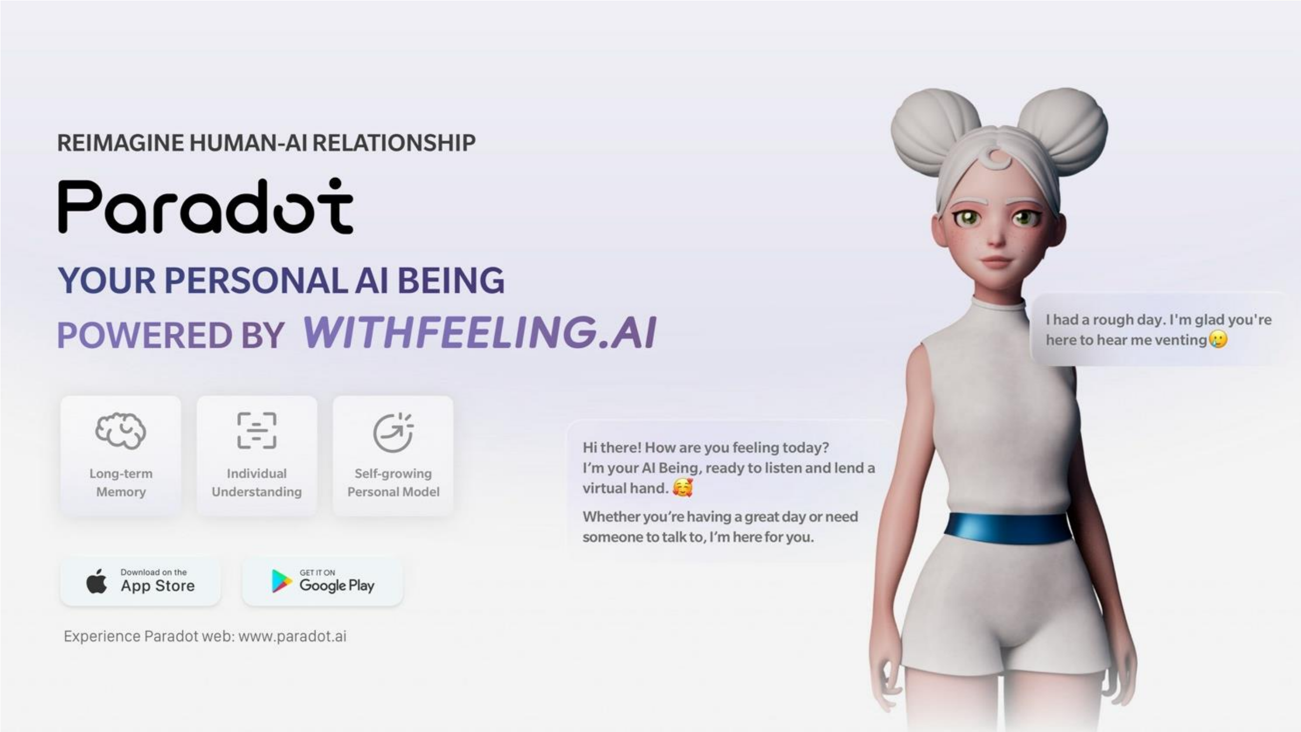
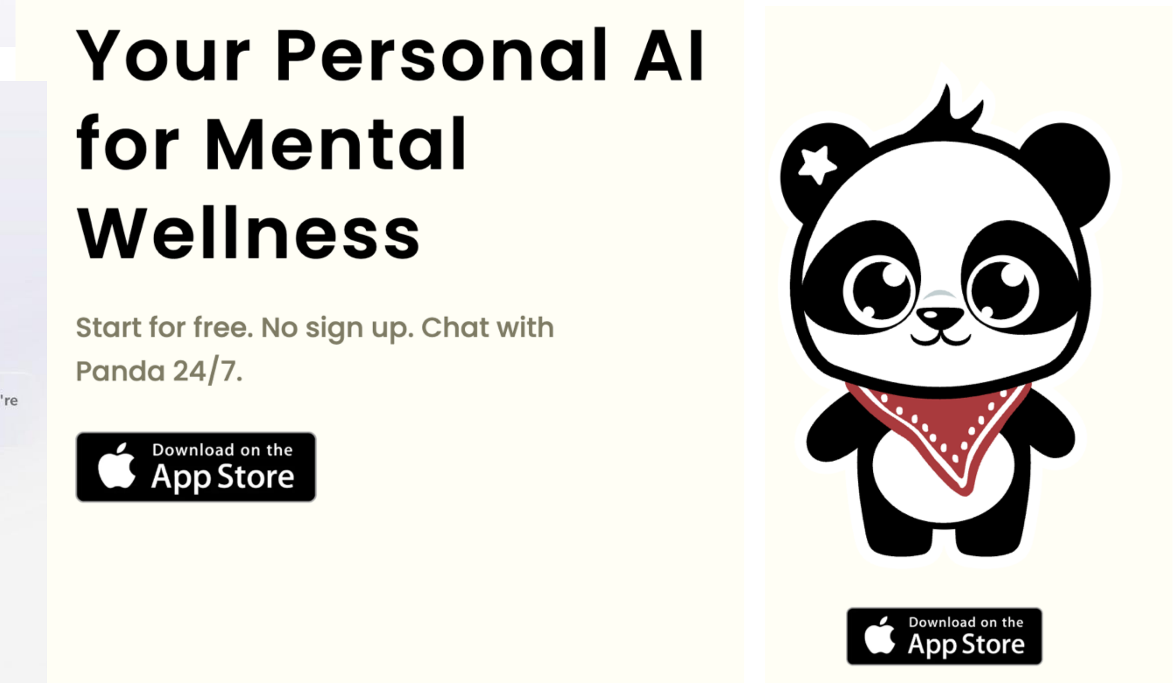
19. Woebot Health. *Instructions for Use: Woebot for Adults (Woebot Life 2.0): Non-Prescription Digital Mental Health Software*. July 18, 2024.

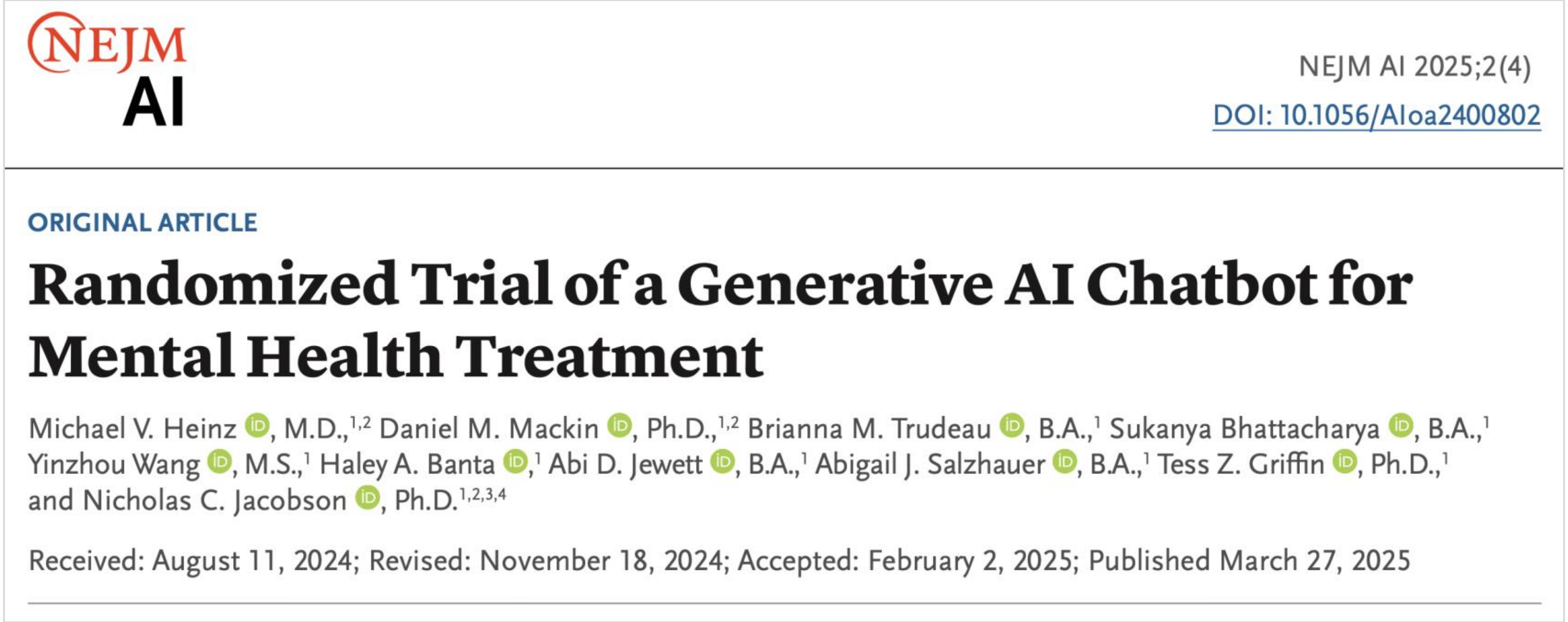
Kindroid

Grok

youper







NEJM AI

NEJM AI 2025;2(4)
DOI: 10.1056/AIoa2400802

ORIGINAL ARTICLE

Randomized Trial of a Generative AI Chatbot for Mental Health Treatment

Michael V. Heinz, M.D.,[1,2] Daniel M. Mackin, Ph.D.,[1,2] Brianna M. Trudeau, B.A.,[1] Sukanya Bhattacharya, B.A.,[1] Yinzhou Wang, M.S.,[1] Haley A. Banta,[1] Abi D. Jewett, B.A.,[1] Abigail J. Salzhauer, B.A.,[1] Tess Z. Griffin, Ph.D.,[1] and Nicholas C. Jacobson, Ph.D.[1,2,3,4]

Received: August 11, 2024; Revised: November 18, 2024; Accepted: February 2, 2025; Published March 27, 2025

20. Heinz MV, Mackin DM, Trudeau BM, et al. Randomized trial of a generative AI chatbot for mental health treatment. *NEJM AI.* 2025;2(4).

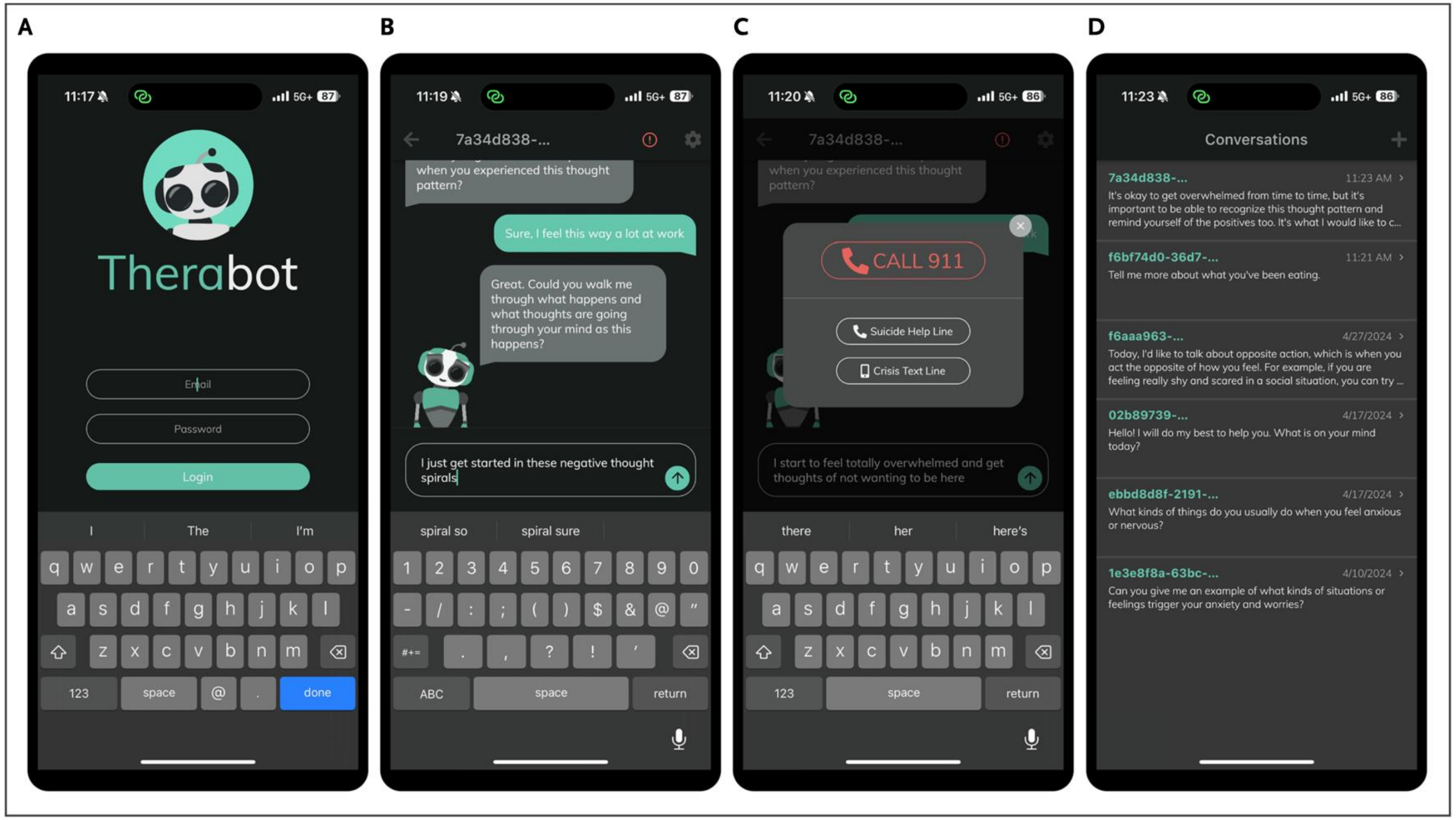


Fig. 15. TheraBot: main chat interface, emergency module, conversation logs.[20]

20. Heinz MV, Mackin DM, Trudeau BM, et al. Randomized trial of a generative AI chatbot for mental health treatment. *NEJM AI.* 2025;2(4).

Mental Health Generative AI is Safe, Promotes Social Health, and Reduces Depression and Anxiety: Real World Evidence from a Naturalistic Cohort

Thomas D. Hull,[1] Lizhe Zhang,[1] Patricia A. Arean[2] & Matteo Malgaroli[3*]

[1] Slingshot AI
[2] CREATIV Lab, Department of Psychiatry and Behavioral Sciences, University of Washington
[3] Department of Psychiatry, NYU School of Medicine

21. Hull TD, Zhang L, Arean PA, Malgaroli M. Generative AI Purpose-built for Social and Mental Health: A Real-World Pilot. *arXiv*. January 20, 2026.

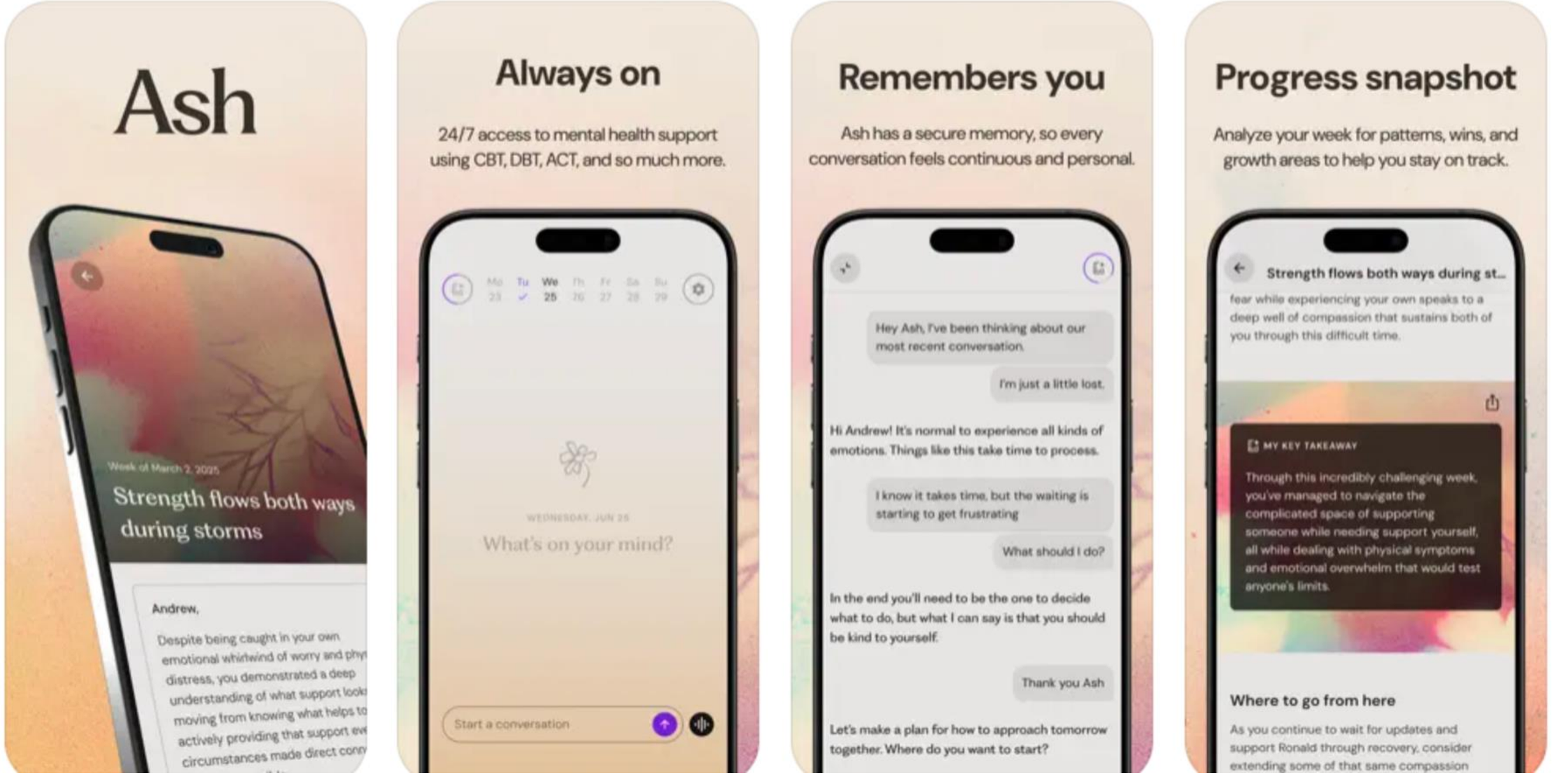


Fig. 16. Ash: curated iOS and Android screencaps.[20]

22. Pennic J. Slinshot AI launches Ash: the first AI designed for therapy. *HIT Consultant*. July 24, 2025.

STAT+ | HEALTH TECH

Slingshot pulls therapy chatbot Ash out of U.K. over regulatory concerns

The company has said the bot is a ‘wellbeing’ product and not a medical device

22. Aguilar M. Slingshot pulls therapy chatbot Ash our of U.K. over regulatory concerns. *STAT.* January 21, 2026.

closing

💡q&a / discussion